



CrateDB Smart Grid Digital Twin Platform

Project Overview

This project represents a next-generation Smart Grid Digital Twin platform designed for modern utility companies and electrical distribution operators. The platform combines real-time IoT streaming, distributed analytics, AI-driven forecasting, SCADA visualization, and digital twin simulation capabilities into a unified architecture.

The objective of the project is to create a scalable and intelligent operational platform capable of:

- Monitoring electrical substations in real time
- Simulating grid behavior and power flow
- Detecting anomalies and predicting failures
- Supporting predictive maintenance strategies
- Providing geo-spatial SCADA visualization
- Enabling AI-driven operational intelligence
- Simulating blackout propagation and load balancing scenarios

The platform was designed around a simulated environment of 1000 substations located in the city of Čakovec, Croatia, each generating real-time telemetry and operational data.

Business Goals

Operational Visibility

Provide complete visibility into the electrical distribution network using real-time telemetry, GIS visualization, and SCADA dashboards.

Predictive Maintenance

Reduce unplanned downtime and maintenance costs by using AI models to predict transformer degradation and equipment failure.

Grid Stability

Detect overloads, voltage instability, harmonics, and abnormal operating conditions before they propagate into larger outages.

Smart Grid Modernization

Enable utility operators to transition toward modern data-driven and AI-assisted Smart Grid operations.

Digital Twin Simulation

Create a virtual representation of the electrical network capable of simulating operational scenarios and future states.

System Architecture

The platform architecture is divided into multiple layers.

1. IoT Simulation Layer

The simulation layer generates real-time telemetry data from virtual substations.

Technology Used

- Locust

Purpose

- Simulate 1000 electrical substations
- Generate real-time sensor data
- Emulate electrical operating conditions
- Simulate overloads, alarms, and abnormal events

Simulated Sensors

Electrical Sensors

- Voltage
- Current
- Frequency
- Active Power
- Reactive Power
- Harmonics (THD)

Thermal Sensors

- Oil Temperature
- Winding Temperature
- Busbar Temperature
- Ambient Temperature

Oil & Gas Monitoring

- Oil Level
- Oil Pressure
- Humidity
- Hydrogen
- Methane
- Acetylene

Operational Alarms

- Overload
- Overheating

Geo-Spatial Information

- GPS coordinates for each substation
 - GIS mapping integration
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2. IoT Messaging & Streaming Layer

The messaging layer is responsible for real-time ingestion and routing of IoT telemetry.

Technology Used

- EMQX
- MQTT protocol

Purpose

- Handle large-scale MQTT device connectivity
- Route sensor telemetry streams
- Manage real-time message delivery
- Support low-latency SCADA communication
- Integrate with analytics and storage layers

Capabilities

- MQTT topic routing
 - Real-time streaming
 - Device session management
 - Rule engine processing
 - High-throughput ingestion
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3. Data Storage & Analytics Layer

The platform uses CrateDB as the primary distributed analytics database.

Technology Used

- CrateDB

Purpose

- Store real-time time-series telemetry
- Provide distributed SQL analytics
- Enable geo-spatial queries
- Support AI feature engineering

- Maintain historical operational state

Key Features

- Distributed SQL architecture
- Time-series optimization
- Native JSON document support
- Geo-spatial indexing
- Real-time analytics
- Horizontal scalability
- PostgreSQL wire protocol compatibility

Stored Data

- Sensor telemetry
- Alarm history
- Geo-location data
- AI scoring data
- Health scores
- Forecasting results
- Historical operational state

4. Backend API & Realtime Layer

The backend layer provides APIs, real-time streaming, and orchestration.

Technologies Used

- FastAPI
- WebSockets
- Socket.IO

Purpose

- Expose REST APIs
- Stream live telemetry to dashboards
- Serve AI prediction data
- Integrate Digital Twin services
- Provide operational APIs for SCADA systems

Core Services

- Real-time station data
 - Alarm APIs
 - Health score APIs
 - AI forecasting APIs
 - Predictive maintenance APIs
 - WebSocket streaming
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5. SCADA Visualization Layer

The visualization layer provides the operational user interface.

Technologies Used

- Vue.js
- Quasar Framework
- Leaflet
- ECharts

Purpose

- Display real-time grid status
- Provide geo-spatial SCADA visualization
- Visualize alarms and health scores
- Show AI predictions and forecasting
- Support operator decision-making

Dashboard Features

- Real-time substation map
 - Alarm management panel
 - Transformer health monitoring
 - Live sensor visualization
 - AI analytics panels
 - Predictive maintenance insights
 - Operational heatmaps
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6. Artificial Intelligence Layer

The AI layer provides predictive analytics and intelligent operational insights.

Technologies Used

- Scikit-learn
- Python AI services

AI Functions

Anomaly Detection

Detect abnormal operating conditions using machine learning models.

Examples

- Overheating detection
- Overload detection
- Harmonics anomalies
- Gas accumulation abnormalities
- Voltage instability

Predictive Maintenance

Estimate equipment degradation and maintenance requirements.

Outputs

- Transformer health score
- Remaining lifetime estimation
- Degradation analysis
- Failure risk scoring

Load Forecasting

Predict future electrical load conditions.

Outputs

- Future current load
- Power demand trends
- Capacity planning indicators

Blackout Prediction

Estimate the probability of outages and cascading failures.

Outputs

- Blackout probability
- Cascade risk analysis
- Critical node identification

7. Digital Twin Layer

The Digital Twin layer creates a virtual representation of the electrical distribution network.

Purpose

- Mirror the real-time operational state
- Simulate future operating conditions
- Model degradation and stress
- Simulate power flow behavior
- Support what-if analysis

Capabilities

- Real-time state synchronization
- Transformer state modeling
- Grid topology representation
- Failure propagation simulation
- Scenario analysis

8. Physics Simulation Layer

The platform includes a simplified power systems simulation engine.

Functions

- Load flow simulation
- Voltage propagation
- Power distribution modeling
- Thermal stress calculation
- Grid overload simulation
- Cascade failure simulation

Engineering Concepts

- Power flow equations
 - Transformer thermal behavior
 - Electrical load balancing
 - Voltage stability analysis
 - Grid stress modeling
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9. Graph Neural Network Layer

The platform includes an advanced AI graph-based model for utility intelligence.

Technologies Used

- PyTorch Geometric
- Graph Neural Networks (GNN)

Purpose

- Model substations as interconnected graph nodes
- Learn relationships between substations
- Predict cascading failures
- Predict load distribution changes
- Improve blackout prediction accuracy

GNN Inputs

- Electrical telemetry
- Thermal data
- Gas monitoring
- Grid topology
- Edge connectivity

GNN Outputs

- Failure probability

- Cascade risk
 - Load prediction
 - Grid stress propagation
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Security & Scalability

Scalability

The architecture is designed for horizontal scaling and large-scale IoT deployments.

Supported Capabilities

- Thousands of substations
- Millions of telemetry records
- Real-time analytics
- Distributed storage
- Multi-user SCADA access

Deployment Readiness

The architecture can be extended toward:

- Kubernetes deployments
 - High-availability clusters
 - Multi-region deployments
 - Edge computing architectures
 - Enterprise observability stacks
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Key Benefits for Utility Operators

Operational Efficiency

- Faster incident response
- Improved operational visibility
- Reduced manual diagnostics

Reliability

- Reduced outage probability
- Faster anomaly detection
- Improved grid stability

Maintenance Optimization

- Predictive maintenance scheduling
- Extended transformer lifetime
- Reduced emergency repairs

AI-Driven Decision Support

- Forecasting and risk analysis
 - Automated operational insights
 - Data-driven planning
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Conclusion

The CrateDB Smart Grid Digital Twin Platform demonstrates how modern distributed analytics, IoT streaming, AI forecasting, and digital twin technologies can be combined into a unified utility operations platform.

The solution provides a foundation for:

- Next-generation SCADA systems
- Smart Grid modernization
- AI-assisted utility operations
- Predictive maintenance strategies
- Real-time operational intelligence
- Digital Twin infrastructure management

By combining CrateDB's distributed analytics capabilities with real-time IoT ingestion and advanced AI modeling, the platform enables utility operators to transition from reactive operations toward predictive and intelligent grid management.